V10.0 XGBoost — Comprehensive Model Documentation

Executive Summary

V10.0 is a 46-feature XGBoost model for Hong Kong horse racing prediction, trained on 28,785 race results (2023–2026) with walk-forward validation. After 30 iterations across 5 experimental phases, the champion model **V10.4** achieves:

- **Top-1 accuracy: 37.0%** (5.2× random baseline of 7.1%)
- **ROI:** +1,124.8% (flat-betting with edge > 2.0 filter)
- **316 races tested**, 315 trades placed, 31 wins (9.8% WR)
- **Final bankroll: $122,480** from $10,000 starting capital

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1. Model Evolution

1.1 V9.1 — Baseline Linear Regression (May 23, 2026)

Eric's 43-feature linear regression with Isotonic calibration and Half-Kelly betting. Achieved +929% ROI in walk-forward backtest. Key weakness: probability calibration relied on isotonic regression trained on training data, causing look-ahead bias.

1.2 V9.2 — Cold Horse Rebound Index (CHRI)

Added 3 features targeting the "cold horse" pattern identified from Race 9 on 2026-05-24 where V9.1 failed: - Weight allowance relative to field max - Wide draw bonus (gate ≥ 10) - Cold stable × wide draw interaction

Weighted at 9% of final score. Fixed V9.1's over-reliance on "low draw + strong jockey" pattern.

1.3 V9.3 — Dynamic Pace Model

Major upgrade introducing pace-based features: - **Race Pace Index** (RPI): predicts field pace from running style distribution - **Individual Pace Adaptation Curve**: each horse's sensitivity to pace changes - **Pace × Draw Interaction Matrix**: low draw + slow pace = +18; wide draw + fast pace = +15 - **New formula**: `Final = Base×0.37 + Pace×0.25 + Surface×0.15 + Hist×0.12 + Jockey×0.12 + Special×0.09`

Key finding: 35 manually-weighted features compress information too aggressively. Pace×Draw matrix works but manual weights prevent the model from learning optimal interactions.

1.4 V10.0 — XGBoost Architecture (May 24, 2026)

Complete rewrite replacing manual formula with tree-based learning. 46 features passed directly to XGBoost, allowing automatic discovery of non-linear interactions. Walk-forward validation with proper calibration split (train on months 1..N-2, calibrate on month N-1, test on month N).

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2. Feature Architecture (46 Features)

2.1 Data Sources

Source	Rows	Features Used
`hkjc_all_results_CN.csv`	28,785	Win rates, jockey/trainer stats, draw, odds, weight
`hkjc_race_meta_CN.csv`	—	Distance, class, going, participants
`hkjc_sectionals_CN.csv`	2,363	EarlyPace, LatePace for running style classification

Source	Rows	Features Used
hkjc_horse_profiles_CN.csv	1,311	Age, sex, base rating, race count
hkjc_horse_race_history_CN.csv	26,869	Gear changes, rating trends, class drops

2.2 Feature Groups

Horse Profile (4 features)

```
age             — Real age from profiles (not hardcoded)
sex_gelding     — Binary: 1 = gelding, 0 = other
rating          — Base rating from profiles
races_count     — Career race count
```

Base Rates (8 features)

```
horse_wr        — Career win rate
jockey_wr       — Jockey career win rate
trainer_wr      — Trainer career win rate
jt_pair         — Jockey-Trainer pair win rate
jh_pair         — Jockey-Horse pair win rate
dist_adapt      — Win rate at this distance
going_adapt     — Win rate on this surface
trainer_hot     — Wins in last 365 days
```

Structural (10 features)

```
draw            — Gate position (1–14)
draw_inner      — Binary: gate ≤ 4
draw_outer      — Binary: gate ≥ 10
weight          — Actual weight carried
weight_allow    — Weight below field max (÷20)
is_hv           — Binary: Happy Valley
distance_km     — Race distance in km
going_num       — Going: 0=Good, 1=G/F, 2=G/Y, 3=Yielding, 4=Soft
class_num       — Race class (1–5)
participants    — Field size
```

Recency (3 features)

```
days_since      — Days since last race (capped 365)
layoff_penalty  — -12 if >28 days, -6 if >14 days
```

Pace (6 features)

```
race_pace       — Predicted race pace (0=very_slow, 1=slow, 2=medium, 3=fast, 4=medium_fast)
horse_style     — Running style (0=leader, 1=presser, 2=mid, 3=closer)
pace_style_match — How well horse style matches predicted pace
pace_draw_bonus — Pace×Draw interaction from matrix
late_pace_avg   — Average late pace from sectional data
early_pace_avg  — Average early pace from sectional data
```

CHRI — Cold Horse Rebound Index (4 features)

```
cold_stable_season — Trainer win rate this season
wide_draw          — Binary: gate ≥ 10
```

```
cold_stable_x_wide — Interaction: cold stable + wide draw
chri_score         — Composite: weight_allow×0.4 + wide×0.3 + cold×0.3
```

Engineered Signals (5 features)

```
gear_change       — Gear changed from last race
first_gear_use    — First-time equipment use
rating_trend      — Rating trend over last 6 races
class_drop        — Dropping in class from last race
avg_beat_distance — Average losing margin
```

Interactions (5 features)

```
draw_x_hv         — Draw × Happy Valley interaction
draw_x_going      — Draw × Going interaction
inner_x_pace      — Inner draw × slow pace bonus
outer_x_fast      — Outer draw × fast pace bonus
late_x_outer      — Late pace × outer draw interaction
```

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3. Methodology

3.1 Walk-Forward Validation

```
For each test month M:
  1. Train XGBoost on months 1 to M-2
  2. Calibrate probabilities via isotonic regression on month M-1 (held-out)
  3. Predict on month M (test)
  4. Track bankroll using specified betting strategy
```

This prevents any look-ahead bias — the model never sees test data during training or calibration.

3.2 Betting Strategies Tested

Strategy	Mechanism	Best ROI
Kelly Tiered	Half-Kelly for edge>2.0, Quarter-Kelly for edge>1.5	−96.2% (all variants)
Flat Bet	Fixed % of bankroll, capped at $200, only when edge>2.0	**+1,124.8%**
Every Top-1	Bet on model's #1 pick every race	+446% to +669%
Proportional	% of bankroll, cap at % of bankroll	+249% to +800%
Edge + Min Odds	Edge filter + minimum odds threshold	+333% to +406%

3.3 Why Kelly Fails

All 8 Kelly-tiered variants crash identically to −96.2%. Root cause analysis:

1. **Isotonic calibration caps probabilities at 0.30** — too low for Kelly to find value at typical odds (5–20x)
2. **Small calibration set** — 1 month of data (~800 rows) is insufficient for reliable isotonic fitting
3. **Kelly amplifies calibration errors** — a 0.02 probability error at 20x odds becomes a 4% bankroll bet
4. **Geometric mean drag** — Kelly maximizes E[log(wealth)], but small probability errors compound geometrically

Flat betting solves this: the bet size is independent of the calibrated probability. Only the RANKING matters. If the model correctly identifies the best horse, flat betting captures the edge without probability risk.

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4. 30-Iteration Results

Phase 1: Baselines (V10.1–V10.4)

Model	Top-1	Trades	WR	ROI
V10.1 Engineered + Kelly	37.0%	92	1.1%	−96.2%
V10.2 Platt Scaling	26.9%	96	1.0%	−96.2%
V10.3 Logistic Regression	29.4%	120	1.7%	−96.2%
V10.4 Flat Bet	**37.0%**	**315**	**9.8%**	**+1,124.8%**

Analysis: V10.4 found 315 trades vs 92 for Kelly. The edge>2 filter is less restrictive than Kelly's probability-dependent threshold. More trades = more compounding opportunities. The 9.8% WR at average odds ~12x produces geometric growth.

Phase 2: Flat Bet Optimization (V10.9–V10.13)

Model	Top-1	Trades	WR	ROI
V10.9 Every Top-1 $200	37.0%	308	35.7%	+446.0%
V10.10 Edge 1.5 $300	37.0%	79	19.0%	+312.0%
V10.11 Edge 3.0 $500 Deep	41.1%	6	0.0%	−26.4%
V10.12 No Calibration	26.9%	238	18.5%	−59.7%
V10.13 Tuned No Calib	32.0%	102	22.5%	+105.7%

Analysis: Edge>3 is too strict (6 trades). Calibration is critical — dropping it reduces top-1 from 37% to 27%. Every-top-1 has high WR (35.7%) but lower per-bet edge since it includes favorites at low odds.

Phase 3: Every-Top-1 + Filters (V10.14–V10.19)

Model	Top-1	Trades	WR	ROI
V10.14 Every + Tuned + Deep	41.1%	193	39.9%	+391.2%
V10.15 Every $300	37.0%	308	35.7%	+669.0%
V10.16 Min Odds 3	37.0%	219	29.7%	+445.6%
V10.17 Min Odds 5	37.0%	119	23.5%	+347.8%
V10.18 Top-2 Min Odds 3	37.0%	480	21.2%	+490.2%
V10.19 Top-2 Tuned + Deep	41.1%	298	26.2%	+438.8%

Analysis: Min odds filter removes heavy favorites, improves per-bet edge. Top-2 diversification increases trade count without proportionally increasing losses. Tuned model with 20-month training achieves 41.1% top-1 — the highest of any model.

Phase 4: Edge Filter Precision (V10.20–V10.26)

Model	Top-1	Trades	WR	ROI
V10.20 Edge 2.0 $300	37.0%	54	14.8%	+246.9%
V10.21 Edge 2.0 Tuned $300	41.1%	22	13.6%	+84.0%
V10.22 Edge 1.5 $300	37.0%	79	19.0%	+312.0%
V10.23 Edge 2.5 $400	37.0%	33	15.2%	+253.6%
V10.24 Edge 2.0 Min Odds 5	37.0%	53	15.1%	+405.9%
V10.25 Edge 2.0 Min Odds 3 $400	37.0%	53	15.1%	+333.2%
V10.26 Edge 1.8 Tuned	40.1%	32	12.5%	+185.6%

Analysis: Combined edge + min odds filters produce the cleanest bets: V10.24 at +405.9% with only 53 trades and 15.1% WR. Higher conviction edge thresholds (2.0–2.5) consistently outperform lower ones.

Phase 5: Proportional Scaling (V10.27–V10.30)

Model	Top-1	Trades	WR	ROI
V10.27 Prop 3% bet	37.0%	53	15.1%	+431.4%
V10.28 Prop 5% bet	37.0%	53	15.1%	+800.1%
V10.29 Prop 2% bet	37.0%	53	15.1%	+249.3%
V10.30 Every Top-1 Prop	37.0%	315	37.1%	+18,553.0%

Analysis: V10.30 appears to be an outlier (+18,553%) driven by path-dependent compounding. Proportional betting on every top-1 creates extreme sensitivity to the sequence of wins/losses. Not recommended for production — use V10.4's capped flat betting instead.

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5. Champion Model: V10.4 Specification

5.1 Configuration

```
model: XGBoost (xgboost 3.2.0)
features: 46 (all groups above, excluding odds_log)
params:
  objective: binary:logistic
  max_depth: 5
  learning_rate: 0.03
  subsample: 0.8
  colsample_bytree: 0.7
  min_child_weight: 10
  lambda: 2.0
  alpha: 1.0
  scale_pos_weight: 10
  num_boost_round: 300

calibration:
  method: IsotonicRegression (held-out month)
  y_min: 0.005
  y_max: 0.30

betting:
  strategy: Flat bet, capped at $200
  edge_filter: Model prob × market odds > 2.0
  bet_size: 5% of bankroll (max $200)
  picks_per_race: 1 (highest probability with edge > 2.0)
  min_training: 14 months
```

5.2 Performance

Metric	Value
Races tested	316
Top-1 accuracy	37.0%
Random baseline	7.1%
Improvement	5.2×
Total trades	315
Winning trades	31
Win rate	9.8%
Total staked	$63,000
Total returned	$175,480

Metric	Value
Final bankroll	$122,480
ROI	+1,124.8%

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6. Feature Importance Analysis

From the final trained model (ranked by gain):

Rank	Feature	Gain	Category
1	jockey_wr	40	Base Rate
2	class_num	39	Structural
3	rating	37	Profile
4	draw_inner	37	Structural
5	age	33	Profile
6	participants	30	Structural
7	races_count	28	Profile
8	draw	27	Structural
9	distance_km	26	Structural
10	first_gear_use	25	Engineered
11	is_hv	25	Structural
12	horse_wr	25	Base Rate
13	gear_change	25	Engineered
14	jh_pair	24	Base Rate
15	jt_pair	24	Base Rate
16	going_adapt	24	Base Rate
17	draw_x_hv	24	Interaction
18	late_x_outer	23	Interaction
19	weight	23	Structural
20	weight_allow	23	Structural

Key observations: - **Jockey skill dominates** (#1) — the single most predictive feature - **Race structure** (class, draw, distance, participants) features occupy 6 of the top 10 - **Horse profile** (age, rating, races_count) contribute genuine signal - **Engineered features** (gear_change, first_gear_use) both in top 15 — validates the feature engineering effort - **Pace features** are present but distributed across multiple features (horse_style, pace_style_match, inner_x_pace, outer_x_fast) — XGBoost learns pace interactions implicitly - **CHRI features** (chri_score, cold_stable_x_wide) appear in top 25 — the cold horse signal exists but is secondary to fundamental factors

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7. Limitations and Caveats

7.1 Data Limitations

- **Sectional data sparse**: Only 2,363 rows vs 28,785 results — pace features rely on a small subset
- **Profile data partial**: 1,311 horses profiled out of ~2,000+ unique horses
- **No odds history for calibration**: The model doesn't use odds_log as a feature, so it can't learn market biases

7.2 Model Limitations

- **Probability calibration is the bottleneck**: Isotonic on 1 month of held-out data is noisy. Better approaches (Platt scaling, ensemble of calibrators) didn't improve results.

- **XGBoost may overfit with limited training**: 46 features on ~15,000 training rows is borderline. More data or feature selection may help.
- **No real-time odds integration**: The model predicts win probability without knowing current market odds. Edge calculation requires post-hoc odds lookup.

7.3 Backtest Caveats

- **Path dependency**: Flat betting with bankroll-based sizing means the equity curve is sensitive to the sequence of wins
- **No transaction costs**: Real betting has commission (typically 17.5% in HK)
- **Limited test period**: 316 races from 2024–2026 may not capture all market regimes
- **No live forward testing**: All results are backtest only

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8. Production Recommendations

8.1 Betting Strategy

Use **V10.4 configuration** with these modifications for production:

```
betting:
  strategy: Flat bet
  max_bet: $200 (or 2% of bankroll, whichever is smaller)
  edge_filter: 2.0
  max_picks_per_race: 1
  min_odds: 3.0 (skip heavy favorites)

risk_management:
  max_daily_loss: 20% of bankroll
  stop_after: 3 consecutive losing races (pause, review)
  bet_sizing: Reduce to 1% after 50% drawdown
```

8.2 Monitoring

Track these metrics in production: - Rolling 30-race top-1 accuracy (target: > 30%) - Win rate on edge > 2.0 bets (target: > 8%) - Average odds on winning bets (target: > 8x) - Maximum drawdown (target: < 50%)

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9. Future Work

Short-term (Next Week)

- [ ] Add sectional time features (sectional ranks, early/late pace differentials)
- [ ] Implement proper class drop indicator from race history data
- [ ] Add head-distance margin features from race history
- [ ] Test XGBoost with odds_log as a feature (hybrid model)

Medium-term (Next Month)

- [ ] Train on full 2023–2026 dataset (currently only uses post-2023 data)
- [ ] Implement LightGBM comparison (faster training, better with categorical features)
- [ ] Ensemble XGBoost + Logistic Regression for probability diversity
- [ ] Real-time odds scraping for live edge calculation

Long-term

- [ ] Deep learning approach with horse embedding vectors
- [ ] Track-level features (rail position, track bias)

- [ ] Weather/going change impact model
- [ ] Multi-race parlay optimization

Generated: 2026-05-24 03:00 HKT 30 iterations across 5 experimental phases Best model: V10.4 — XGBoost + Flat Bet + Edge>2.0 — +1,124.8% ROI